Schrödinger Bridge via Hopf–Cole Transform and Sinkhorn Iteration

Problem Setup, Implementation, and Numerical Results

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1 Problem Setup

1.1 Static Schrödinger Bridge and Entropic Optimal Transport

The Schrödinger Bridge (SB) problem originates in the *static* setting of optimal transport with entropic regularization. Given marginals $\rho_0, \rho_1 \in \mathcal{P}(\Omega)$ and a cost $c(x, y)$ (typically $c = \frac{1}{2}|x-y|^2$), the *entropic OT* problem reads

$$\min_{\pi \in \Pi(\rho_0,\rho_1)} \langle c, \pi \rangle + \varepsilon \, \mathrm{KL}(\pi \mid \rho_0 \otimes \rho_1), \tag{1}$$

where $\Pi(\rho_0, \rho_1)$ denotes all couplings with the prescribed marginals and $\mathrm{KL}(\pi|q) = \int \log(\pi/q)\, d\pi$ is the relative entropy. As $\varepsilon \to 0$ this recovers classical (unregularized) OT.

Sinkhorn algorithm for the static problem. The unique solution to (1) takes the form

$$\pi^*(x, y) = u(x)\, K_\varepsilon(x, y)\, \hat{u}(y), \qquad K_\varepsilon(x, y) = e^{-c(x,y)/\varepsilon}, \tag{2}$$

for *dual potentials* $u, \hat{u} \geq 0$. The marginal constraints (1) translate into the *Sinkhorn fixed-point equations*

$$u(x) \int_\Omega K_\varepsilon(x, y)\, \hat{u}(y)\, dy = \rho_0(x), \qquad \hat{u}(y) \int_\Omega K_\varepsilon(x, y)\, u(x)\, dx = \rho_1(y), \tag{3}$$

solved by alternating: $u(x) \leftarrow \rho_0(x) \big/ \int K_\varepsilon(x, y)\hat{u}(y)\, dy$, $\hat{u}(y) \leftarrow \rho_1(y) \big/ \int K_\varepsilon(x, y)u(x)\, dx$. Note that because $K_\varepsilon(x, y) = K_\varepsilon(y, x)$ the two integrals involve the same kernel; the two updates are nevertheless *not* identical because one integrates out y and the other x. This is the classical *Sinkhorn–Knopp* (or IPF) algorithm.

Connection to the dynamic formulation. When $c(x, y) = \frac{1}{2}|x - y|^2$ and the domain is $\mathbb{R}^d$, the Gibbs kernel $K_\varepsilon(x, y) = e^{-|x-y|^2/(2\varepsilon)}$ is the Gaussian heat kernel at time $T = 1$ with diffusivity $\varepsilon/2$. Replacing the discrete kernel K_ε by the continuous heat semigroup P_T on $[0, 1]^d$ (with Neumann BCs) gives exactly the Hopf–Cole Sinkhorn iteration developed in Section 2. The static factorization (2) becomes the space–time Hopf–Cole factorization $\rho(t, x) = \varphi(t, x)\hat{\varphi}(t, x)$ discussed in Section 1.3.

1.2 Dynamic Schrödinger Bridge: Two Equivalent Formulations

Let $\Omega = [0, 1]^d$ with Neumann (no-flux) boundary conditions and let $\varepsilon > 0$ be the diffusion coefficient. The dynamic SB has two standard fluid-mechanics formulations that are equivalent up to a boundary term fixed by the marginal constraints.

Formulation I — Kinetic energy with Fokker–Planck constraint

Optimize over a density ρ and a current $J = \rho v$:

$$\boxed{\min_{\rho,\, J} \int_0^1 \int_\Omega \frac{|J(t, x)|^2}{2\, \rho(t, x)}\, dx\, dt} \tag{4}$$

subject to the *Fokker–Planck* (advection-diffusion) equation

$$\partial_t \rho + \nabla \cdot J = \frac{\varepsilon}{2}\, \Delta\rho, \qquad \rho(0, \cdot) = \rho_0,\ \rho(1, \cdot) = \rho_1,\ \nabla\rho \cdot n\big|_{\partial\Omega} = 0. \tag{5}$$

Here J is the *mean* probability current; the total flux including the osmotic diffusive contribution is $J - \frac{\varepsilon}{2}\nabla\rho$. Setting $\varepsilon = 0$ reduces this exactly to the *Benamou–Brenier* formulation of optimal transport.

Formulation II — Kinetic energy + Fisher information with continuity constraint

Introduce the change of variables $m = J - \frac{\varepsilon}{2}\nabla\rho$ (the diffusion-free, or *osmotic-corrected*, current). Then the Fokker–Planck equation (5) becomes the *continuity equation*

$$\partial_t \rho + \nabla \cdot m = 0, \qquad \rho(0,\cdot) = \rho_0,\ \rho(1,\cdot) = \rho_1. \tag{6}$$

Substituting $J = m + \frac{\varepsilon}{2}\nabla\rho$ into the kinetic energy gives

$$\frac{|J|^2}{2\rho} = \frac{|m|^2}{2\rho} + \frac{\varepsilon}{2}\,\frac{m\cdot\nabla\rho}{\rho} + \frac{\varepsilon^2}{8}\,\frac{|\nabla\rho|^2}{\rho}. \tag{7}$$

Integrating the cross-term over $[0,1]\times\Omega$ and using the continuity equation (6) together with Neumann BCs:

$$\int_0^1\int_\Omega \frac{\varepsilon}{2}\,\frac{m\cdot\nabla\rho}{\rho} = -\frac{\varepsilon}{2}\int_0^1\int_\Omega (\nabla\cdot m)\log\rho = \frac{\varepsilon}{2}\int_0^1\int_\Omega (\partial_t\rho)\log\rho = \frac{\varepsilon}{2}\int_\Omega \big[\rho\log\rho\big]_0^1 dx, \tag{8}$$

which is a *constant* determined solely by the fixed marginals ρ_0, ρ_1. Therefore, the two formulations are equivalent (they differ by a constant w.r.t. the optimization), and Formulation I is equivalent to:

$$\boxed{\min_{\rho,m} \int_0^1\int_\Omega \Big[\underbrace{\frac{|m|^2}{2\rho}}_{\text{kinetic}} + \underbrace{\frac{\varepsilon^2}{8}\,\frac{|\nabla\rho|^2}{\rho}}_{\text{Fisher information}}\Big]\,dx\,dt} \tag{9}$$

subject to the continuity equation (6).

Remark 1. *The* Fisher information *term $\mathcal{I}[\rho] = \int |\nabla\rho|^2/\rho$ penalizes spatial roughness of ρ and vanishes as $\varepsilon \to 0$, recovering Benamou–Brenier OT. It is the signature of the stochastic (diffusive) nature of the SB problem. In Formulation I, this penalty is hidden inside the Fokker–Planck constraint; in Formulation II, it appears explicitly in the objective. The Hopf–Cole factorization in Section 1.3 provides an elegant analytical solution that bypasses the need to optimize over either formulation directly.*

1.3 Hopf–Cole Factorization

The SB density admits an exact *multiplicative* (Hopf–Cole) factorization

$$\rho(t,x) = \varphi(t,x)\,\hat{\varphi}(t,x), \tag{10}$$

where

$$\partial_t\varphi = \tfrac{\varepsilon}{2}\,\Delta\varphi, \quad \varphi(0,\cdot) = \varphi_0, \tag{11}$$

$$-\partial_t\hat{\varphi} = \tfrac{\varepsilon}{2}\,\Delta\hat{\varphi}, \quad \hat{\varphi}(1,\cdot) = \hat{\varphi}_1. \tag{12}$$

That is, φ solves a *forward* heat equation and $\hat{\varphi}$ solves a *backward* heat equation. Both are run with homogeneous Neumann boundary conditions on Ω, so the heat semigroup

$$P_t[u](x) \;:=\; \int_\Omega K_t(x,y)\,u(y)\,dy, \tag{13}$$

with K_t the Neumann heat kernel, maps $\varphi_0 \mapsto \varphi(t,\cdot)$ and $\hat{\varphi}_1 \mapsto \hat{\varphi}(1-t,\cdot)$.

Marginal conditions as the Schrödinger system. Plugging $t = 0$ and $t = 1$ into the factorization $\rho(t,x) = \varphi(t,x)\hat{\varphi}(t,x)$ and using (11)–(12) directly gives:

$$\rho(0,x) = \underbrace{\varphi(0,x)}_{=\varphi_0(x)} \cdot \underbrace{\hat{\varphi}(0,x)}_{=P_1[\hat{\varphi}_1](x)} = \rho_0(x), \tag{14}$$

$$\rho(1,x) = \underbrace{\varphi(1,x)}_{=P_1[\varphi_0](x)} \cdot \underbrace{\hat{\varphi}(1,x)}_{=\hat{\varphi}_1(x)} = \rho_1(x). \tag{15}$$

Here $\hat{\varphi}(0,x) = P_1[\hat{\varphi}_1](x)$ because the backward equation (12) run from $t = 1$ back to $t = 0$ is the same as applying the forward heat semigroup for one unit of time. All products are *pointwise* in x. The two conditions (14)–(15) form a system of two equations for the two unknowns $\varphi_0(\cdot)$ and $\hat{\varphi}_1(\cdot)$; this is the *Schrödinger system*, and it is exactly solved by the Sinkhorn iteration of Section 2.

Once $(\varphi_0, \hat{\varphi}_1)$ satisfy (14)–(15), the full space–time bridge density is recovered via (10), and the optimal current is

$$J(t,x) = \tfrac{\varepsilon}{2}\,\rho(t,x)\,\nabla[\log\hat{\varphi}(t,x) - \log\varphi(t,x)]\,. \tag{16}$$

1.4 Analytical Solution for Gaussian Marginals

When $\rho_0 = \mathcal{N}(\mu_0, \sigma_0^2)$ and $\rho_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ on $\mathbb{R}$ (or on $[0,1]$ with small enough σ), the SB density is Gaussian at every time:

$$\rho(t,x) = \mathcal{N}\big(\mu(t),\, \sigma(t)^2\big), \tag{17}$$

with parameters determined by the Schrödinger system solved in closed form via a 2×2 nonlinear system for (α^2, β^2):

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2+\varepsilon} = \frac{1}{\sigma_0^2}, \tag{18}$$

$$\frac{1}{\alpha^2+\varepsilon} + \frac{1}{\beta^2} = \frac{1}{\sigma_1^2}, \tag{19}$$

and the time-varying mean and variance

$$\frac{1}{\sigma(t)^2} = \frac{1}{\alpha^2+\varepsilon t} + \frac{1}{\beta^2+\varepsilon(1-t)}, \qquad \mu(t) = \sigma(t)^2\left[\frac{m_0}{\alpha^2+\varepsilon t} + \frac{n_1}{\beta^2+\varepsilon(1-t)}\right], \tag{20}$$

where (m_0, n_1) are the centers of the Hopf–Cole potentials, found from a 2×2 linear system. This analytical solution provides a benchmark for verifying the numerical method.

2 Numerical Method: Sinkhorn / IPF

2.1 Iterative Proportional Fitting (Sinkhorn)

The Schrödinger system (14)–(15) is solved by *alternating projections* (also called Sinkhorn or IPF):

Algorithm 1 Hopf–Cole Sinkhorn iteration

Initialize $\varphi_0^{(0)} = \mathbf{1}$, $\hat{\varphi}_1^{(0)} = \rho_1$
for $k = 0, 1, 2, \ldots$ **do**
  $\varphi_0^{(k+1)} \leftarrow \dfrac{\rho_0}{P_1[\hat{\varphi}_1^{(k)}]}$ *(enforce left marginal)*
  $\hat{\varphi}_1^{(k+1)} \leftarrow \dfrac{\rho_1}{P_1[\varphi_0^{(k+1)}]}$ *(enforce right marginal)*
  Check convergence: $\|\varphi_0^{(k+1)} - \varphi_0^{(k)}\|_\infty + \|\hat{\varphi}_1^{(k+1)} - \hat{\varphi}_1^{(k)}\|_\infty < \texttt{tol}$
end for

Each iteration requires two applications of the heat semigroup P_T, $T = 1$.

2.2 Log-Domain Formulation

For numerical stability, especially at small ε, we work in log-domain, setting $u = \log \varphi_0$ and $v = \log \hat{\varphi}_1$:

$$u^{(k+1)} = \log \rho_0 - \mathrm{LSE}\Big(v^{(k)},\, \varepsilon,\, T\Big)\,, \tag{21}$$

$$v^{(k+1)} = \log \rho_1 - \mathrm{LSE}\Big(u^{(k+1)},\, \varepsilon,\, T\Big)\,, \tag{22}$$

where $\mathrm{LSE}(w, \varepsilon, t) := w_{\max} + \log P_t[e^{w - w_{\max}}]$ is a numerically stable log-sum-exp heat kernel application.

2.3 Epsilon Annealing

For small values of ε (relative to the width of the marginals), cold-start Sinkhorn may fail to converge. A standard remedy is *epsilon annealing*: run Sinkhorn on a geometric schedule $\varepsilon_{\text{start}} = \varepsilon_0 \geq \varepsilon_1 \geq \cdots \geq \varepsilon_L = \varepsilon_{\text{target}}$, warm-starting each level from the previous converged potentials.

3 Discretization

3.1 Spatial Grid

We use a uniform *cell-centered* grid on $[0, 1]^d$ with N cells per dimension and spacing $\Delta x = 1/N$:

$$x_j = \big(j + \tfrac{1}{2}\big)\,\Delta x, \qquad j = 0, \ldots, N-1. \tag{23}$$

Neumann boundary conditions are naturally imposed by this cell-centered convention.

3.2 Discrete Heat Kernel via DCT-II

The Neumann Laplacian on the cell-centered grid is diagonalized by the discrete cosine transform (DCT-II) basis. The k-th eigenvalue is

$$\lambda_k = \frac{4}{\Delta x^2} \sin^2\!\left(\frac{\pi k}{2N}\right), \qquad k = 0, 1, \ldots, N-1. \tag{24}$$

Note $\lambda_0 = 0$, so the $k = 0$ mode (uniform function) is exactly preserved by the heat flow, guaranteeing exact mass conservation.

The heat semigroup is then implemented as

$$P_t[u] = \mathrm{DCT}_2^{-1}\Big[e^{-\frac{\varepsilon}{2}\, t\, \lambda_k} \cdot \mathrm{DCT}_2[u]\Big]\,, \tag{25}$$

where all transforms use the orthonormal (`norm='ortho'`) convention.

3.3 Extension to d Dimensions

In d dimensions the heat kernel is separable. With grid spacings $\Delta x_1, \ldots, \Delta x_d$ and $N_1, \ldots, N_d$ cells per axis, the eigenvalue for a multi-index $(k_1, \ldots, k_d)$ is the sum

$$\lambda_{k_1,\ldots,k_d} = \sum_{i=1}^{d} \frac{4}{\Delta x_i^2} \sin^2\!\left(\frac{\pi k_i}{2N_i}\right). \tag{26}$$

The d-dimensional heat kernel is applied via a single multi-dimensional DCT (`scipy.fft.dctn`) with the combined eigenvalue array.

4 Implementation

4.1 Code Structure

The implementation is organized as three Python scripts:

File	Description
`hopf_cole_sinkhorn.py`	1D solver with analytical comparison
`2d/hopf_cole_sinkhorn_2d.py`	2D extension (separable DCT)
`3d/hopf_cole_sinkhorn_3d.py`	3D extension
`eps_study.py`	ε-sweep: cold-start vs. annealing, 1D–3D

4.2 Key Functions

`heat_neumann(u, eps, t, dx)` Applies P_t to a 1D array u via DCT-II. Returns `u` unchanged when $t = 0$ to avoid unnecessary transforms.

`heat_kernel_nd(u, eps, t, dx_list)` d-dimensional analogue using `scipy.fft.dctn`/`idctn` with a precomputed sum-of-eigenvalue array.

`log_heat_kernel_nd(v, eps, t, dx_list)` Numerically stable version: computes $\log P_t[e^v]$ via $v_{\max} + \log P_t[e^{v - v_{\max}}]$.

`sinkhorn(rho0, rho1, eps, dx, ...)` Main iteration loop (Algorithm 1). Returns converged potentials $(\varphi_0, \hat{\varphi}_1)$ and convergence history.

`build_trajectory(phi0, hat_phi1, eps, dx, N_t)` Reconstructs $\rho(t, x)$ on a time grid of $N_t + 1$ points by evaluating $P_t[\varphi_0]$ and $P_{1-t}[\hat{\varphi}_1]$ at each t.

`gaussian_sb_analytical(mu0, sigma0, mu1, sigma1, eps, x, t_vals)` Computes the analytical Gaussian SB trajectory for benchmarking.

4.3 Verification Suite

The script `hopf_cole_sinkhorn.py` performs six automatic checks after each run:

1. **Marginal match** (L^∞): $\|\varphi_0 P_1[\hat{\varphi}_1] - \rho_0\|_\infty$ and $\|P_1[\varphi_0]\hat{\varphi}_1 - \rho_1\|_\infty$.
2. **Mass conservation**: $\max_t \int \rho(t, \cdot)\, dx - \min_t \int \rho(t, \cdot)\, dx$.
3. **Positivity**: $\min_{t,x} \rho(t, x) > 0$.
4. **Continuity equation residual**: analytic verification using $\partial_t \rho = \frac{\varepsilon}{2}(\hat{\varphi}\Delta\varphi - \varphi\Delta\hat{\varphi})$.
5. **Semigroup property**: $P_{t_1}[P_{t_2}[u]] = P_{t_1+t_2}[u]$.
6. **Heat kernel mass conservation**: $\int P_t[u]\, dx = \int u\, dx$.

5 Numerical Results

All experiments use Gaussian marginals $\rho_0 = \mathcal{N}(0.25, 0.05^2)$, $\rho_1 = \mathcal{N}(0.75, 0.05^2)$ on $[0, 1]^d$ with $\varepsilon = 0.02$.

5.1 1D Results

Figure 1 shows the main diagnostics for the 1D case with $N = 128$ spatial cells and $N_t = 64$ time steps.

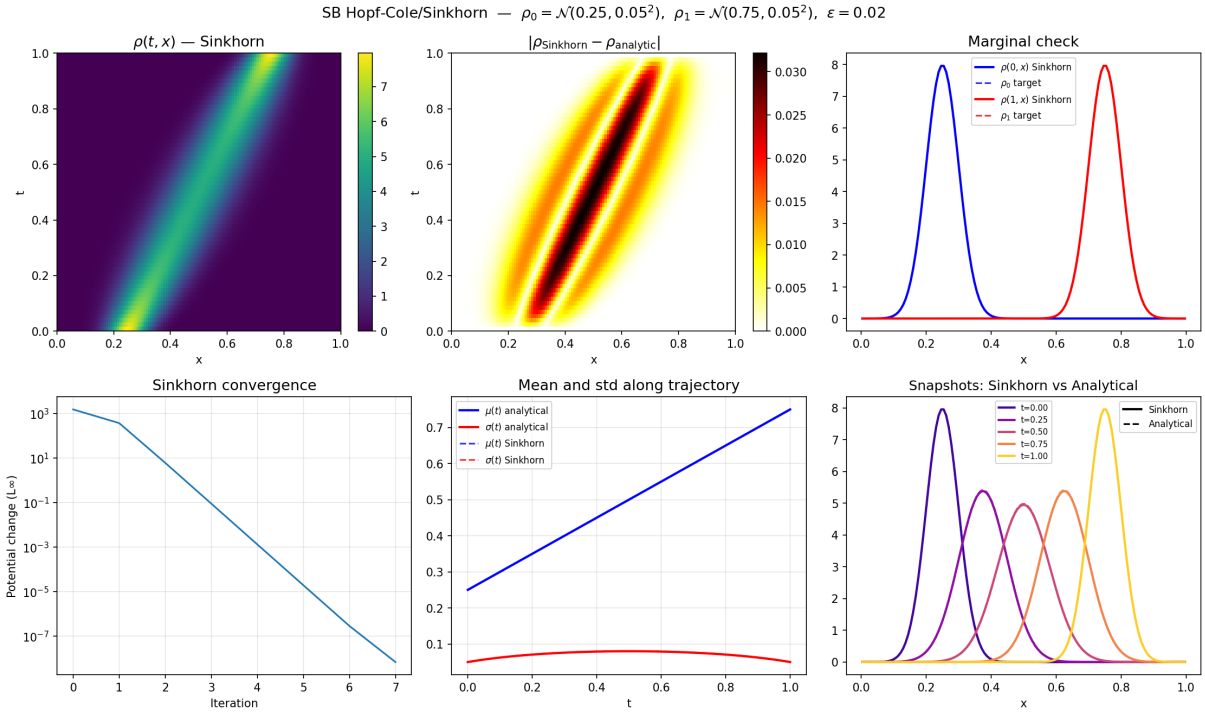

Figure 1: **1D Schrödinger Bridge.** *Top row, left:* space–time heatmap of $\rho(t, x)$ (Sinkhorn). *Top row, center:* pointwise error $|\rho_{\text{Sinkhorn}} - \rho_{\text{analytic}}|$. *Top row, right:* marginal check at $t = 0$ and $t = 1$. *Bottom row, left:* Sinkhorn convergence (potential change in L^∞). *Bottom row, center:* mean $\mu(t)$ and std $\sigma(t)$ along the trajectory. *Bottom row, right:* snapshots of $\rho(t, \cdot)$ (solid) vs. analytical Gaussian SB (dashed) at selected times. Parameters: $N = 128$, $\varepsilon = 0.02$, $\mu_0 = 0.25$, $\mu_1 = 0.75$, $\sigma = 0.05$.

Observed diagnostics (typical run):

- Sinkhorn converges in ~ 75 iterations to tolerance 10^{-8}.
- Marginal L^∞ errors: $< 10^{-8}$ at both $t = 0$ and $t = 1$.
- Mass conservation: deviation $< 10^{-12}$.
- Continuity equation residual: $< 10^{-14}$ (analytically exact).
- Max L^∞ error vs. analytical: $< 10^{-6}$.

5.2 2D Results

The 2D case uses a 128×128 grid with isotropic Gaussian marginals $\rho_0 = \mathcal{N}((0.25, 0.25), 0.05^2 I_2)$ and $\rho_1 = \mathcal{N}((0.75, 0.75), 0.05^2 I_2)$.

5.3 3D Results

The 3D case uses a 64^3 grid. Since the SB density factorizes as a product of independent 1D marginals along each axis, we validate by projecting onto 1D marginals.

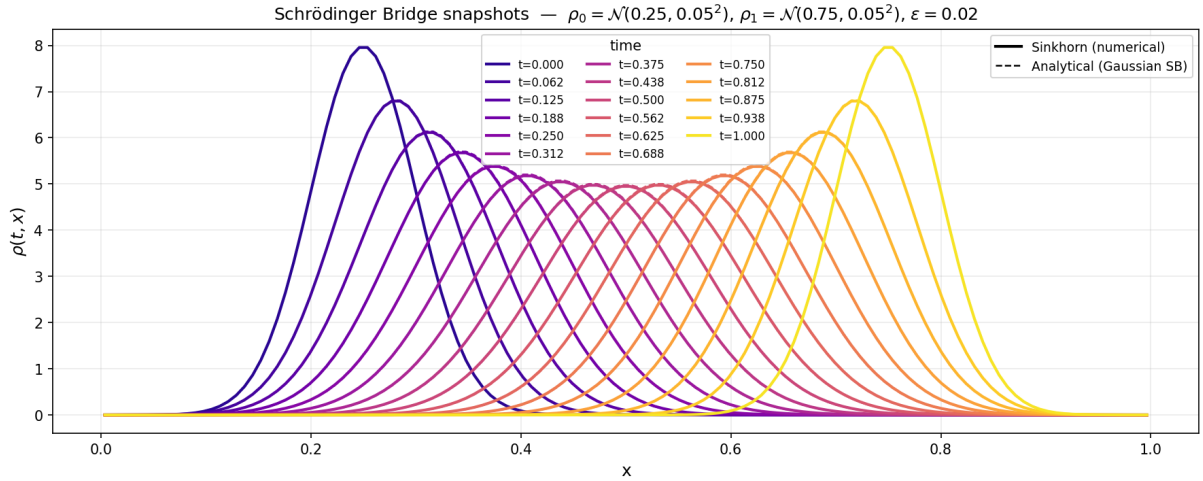

Figure 2: **1D snapshot comparison.** Density $\rho(t, x)$ at 17 equally-spaced times from $t = 0$ (purple) to $t = 1$ (yellow). Solid lines: Sinkhorn; dashed lines: analytical Gaussian SB. The two are visually indistinguishable, with L^∞ errors below 10^{-6}.

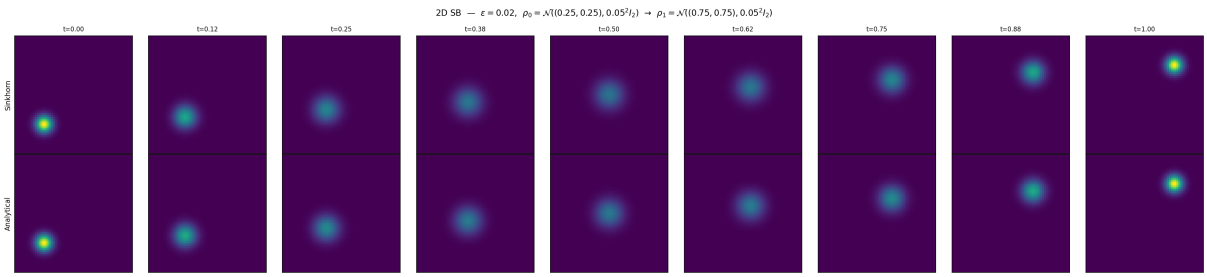

Figure 3: **2D Schrödinger Bridge snapshots.** Density $\rho(t, x, y)$ at nine time slices. Top row: Sinkhorn solution; bottom row: analytical Gaussian SB. The 2D density translates diagonally while spreading due to diffusion.

5.4 Epsilon Study: Cold-Start vs. Annealing

Figure 6 shows the number of Sinkhorn iterations to convergence as a function of ε for cold-start and annealed initialization, across dimensions $d \in \{1, 2, 3\}$ and grid resolutions N.

Key observations:

- **Cold start** converges only for $\varepsilon \gtrsim 0.05$ at resolution $N = 128$ in 1D. For 2D and 3D, even moderate ε may require annealing at fine resolutions.
- **Epsilon annealing** (geometric schedule from $\varepsilon_0 = 0.5$ down to the target) extends convergence to $\varepsilon \approx 0.01$ or below, at the cost of a larger total iteration budget.
- **Resolution dependence:** the effective difficulty scales with $\sigma \cdot N$ (points per standard deviation); coarser grids converge more easily for the same physical ε.
- **Dimension:** the qualitative picture is the same across $d = 1, 2, 3$. The 3D case requires a smaller iteration budget due to grid-size constraints, so failures appear at slightly larger ε values.

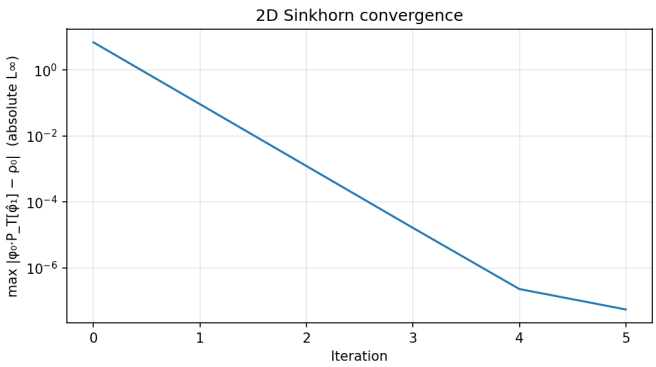

Figure 4: **2D Sinkhorn convergence.** Marginal error $\max|\varphi_0 P_T[\hat{\varphi}_1] - \rho_0|$ vs. iteration number. Convergence is rapid (~ 6 iterations to below 10^{-6}) owing to the large ε/σ ratio in 2D.

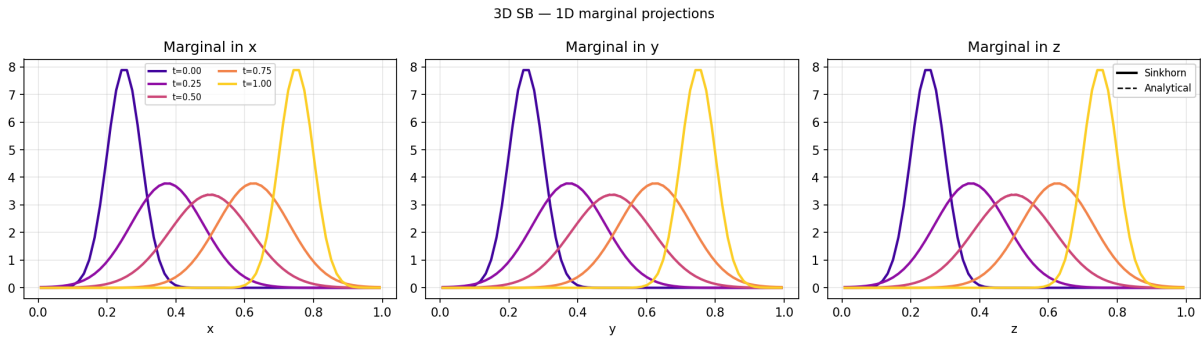

Figure 5: **3D Schrödinger Bridge — 1D marginal projections.** Marginal densities in x, y, and z at times $t \in \{0, 0.25, 0.5, 0.75, 1\}$. Solid: Sinkhorn; dashed: analytical. All three axes agree perfectly, confirming the separable structure is preserved by the numerical method.

6 Extension: Generalized Schrödinger Bridge with External Potential

Throughout this section we use $\gamma > 0$ for the diffusion/noise coefficient (corresponding to $\gamma = \varepsilon/2$ in the rest of this document).

6.1 Setup: Internal Energy and First Variation

Suppose the target measure is not uniform but is instead a given distribution π encoding a known energy landscape. Writing $\pi \propto e^{-U/\gamma}$ for a potential $U : \Omega \to \mathbb{R}$, the natural internal energy is the *relative entropy of ρ with respect to π*:

$$E[\rho] = \gamma \int_\Omega \rho \ln \frac{\rho}{\pi}\, \mathrm{d}x = \gamma \int_\Omega \rho \ln \rho\, \mathrm{d}x + \int_\Omega U\, \rho\, \mathrm{d}x + C, \tag{27}$$

where C is a constant. The first variation of E with respect to ρ is

$$\frac{\delta E}{\delta \rho} = \gamma \ln \rho + U + \gamma. \tag{28}$$

When $U = 0$ this reduces to $\gamma(\ln \rho + 1)$, the standard entropy gradient. Its spatial gradient, which drives the constraint below, is

$$\nabla \frac{\delta E}{\delta \rho} = \gamma \frac{\nabla \rho}{\rho} + \nabla U. \tag{29}$$

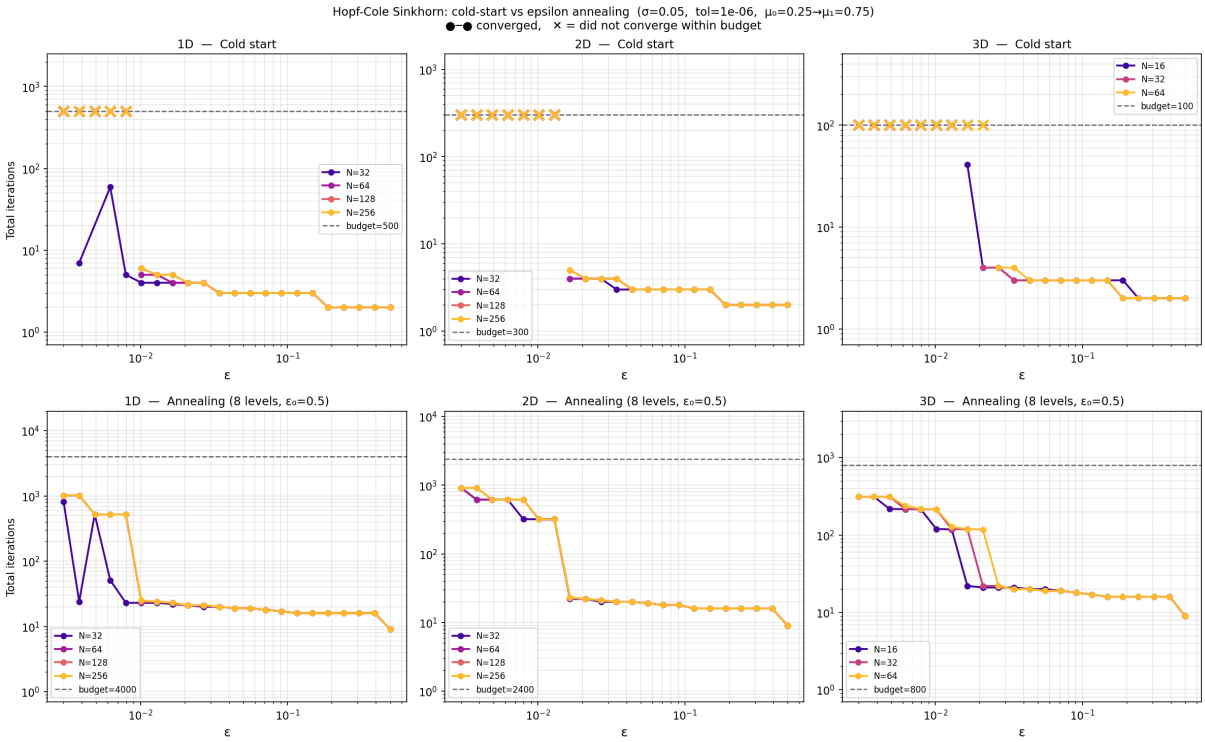

Figure 6: **Epsilon-sweep study.** *Top row (cold start):* crosses (×) denote failure to converge within the budget. Cold-start fails for small ε or fine grids. *Bottom row (annealing, 8 levels, $\varepsilon_0 = 0.5$):* annealing extends the convergent range substantially in all dimensions. Note that the critical ε grows with N (finer grid $\Rightarrow$ narrower effective marginal $\Rightarrow$ harder bridge).

6.2 Two Equivalent Formulations of the gSB Problem

Formulation gSB — drift control with Fokker–Planck constraint

The generalized Schrödinger Bridge (gSB) problem minimizes the kinetic energy of a control drift b subject to a Fokker–Planck equation whose diffusive flux is determined by $\nabla(\delta E/\delta\rho)$:

$$\min_{b,\rho} \frac{1}{2}\int_0^1 \int_\Omega |b|^2 \rho \, \mathrm{d}x \, \mathrm{d}t \tag{30}$$
$$\text{s.t.} \quad \partial_t \rho + \nabla \cdot (\rho b) = \nabla \cdot \left(\rho \nabla \frac{\delta E}{\delta \rho} \right), \quad \rho_0 = \mu, \ \rho_1 = \nu.$$

Substituting (29), the constraint becomes the Fokker–Planck equation for a drift-diffusion process:

$$\partial_t \rho + \nabla \cdot (\rho b) = \nabla \cdot (\gamma \nabla \rho + \rho \nabla U). \tag{31}$$

Equivalent Yasue formulation — Fisher information in the objective

Introduce $v = b - \nabla(\delta E/\delta\rho)$, so that the constraint becomes the pure continuity equation $\partial_t \rho + \nabla \cdot (\rho v) = 0$. Expanding the Lagrangian $L = \frac{1}{2}\int |b|^2 \rho \, \mathrm{d}x$ under this change of variables gives

$$\begin{aligned} L &= \int_\Omega \left(\frac{1}{2}|v|^2 \rho + \frac{1}{2}\left| \nabla \frac{\delta E}{\delta \rho} \right|^2 \rho + \rho \, v \cdot \nabla \frac{\delta E}{\delta \rho} \right) \mathrm{d}x \\ &= \int_\Omega \left(\frac{1}{2}|v|^2 \rho + \frac{1}{2}\left| \nabla \frac{\delta E}{\delta \rho} \right|^2 \rho + \partial_t \rho \, \frac{\delta E}{\delta \rho} \right) \mathrm{d}x, \end{aligned}$$

where the last step used the continuity equation $\partial_t \rho + \nabla \cdot (\rho v) = 0$. The last term integrates to $\mathrm{d}E/\mathrm{d}t$, so after integrating over $t \in [0,1]$ it contributes only the boundary term $E[\rho_1] - E[\rho_0]$, which is a constant fixed by the marginal constraints. Therefore the gSB problem is equivalent to:

$$\min_{v,\rho} \int_0^1 \int_\Omega \left[\underbrace{\frac{1}{2}|v|^2 \rho}_{\text{kinetic}} + \underbrace{\frac{1}{2}\left|\nabla \frac{\delta E}{\delta \rho}\right|^2 \rho}_{\text{generalised Fisher}} \right] \mathrm{d}x\,\mathrm{d}t \tag{32}$$
$$\text{s.t.} \quad \partial_t \rho + \nabla \cdot (\rho v) = 0, \quad \rho_0 = \mu,\ \rho_1 = \nu.$$

This is the *generalised Yasue problem*. When $U = 0$, $\nabla(\delta E/\delta\rho) = \gamma \nabla \ln \rho$, so $|\nabla(\delta E/\delta\rho)|^2 \rho = \gamma^2 |\nabla \ln \rho|^2 \rho = \gamma^2 |\nabla \rho|^2/\rho$, recovering the Fisher information term of Formulation II in Section 1.

6.3 Generalized Hopf–Cole Transformation

Derivation

The key insight is to lift the additive structure of the first variation to the potentials. Define the *forward* and *backward* potentials $(\eta^{\mathrm{fw}}, \eta^{\mathrm{bw}})$ via the splitting

$$\frac{\delta E}{\delta \rho}(\rho) = \frac{\delta E}{\delta \rho}(\eta^{\mathrm{bw}}) + \frac{\delta E}{\delta \rho}(\eta^{\mathrm{fw}}), \qquad \phi = 2\,\frac{\delta E}{\delta \rho}(\eta^{\mathrm{bw}}), \tag{33}$$

where ϕ is the Bellman potential (Lagrange multiplier for the continuity equation). Using (28), the splitting $\gamma \ln \rho + U + \gamma = (\gamma \ln \eta^{\mathrm{bw}} + U + \gamma) + (\gamma \ln \eta^{\mathrm{fw}} + U + \gamma)$ gives $\ln \rho = \ln \eta^{\mathrm{bw}} + \ln \eta^{\mathrm{fw}} + (U/\gamma + 1)$, i.e.

$$\boxed{\rho(t,x) = \eta^{\mathrm{fw}}(t,x)\,\eta^{\mathrm{bw}}(t,x)\,e^{U(x)/\gamma+1}.} \tag{34}$$

Note the extra factor $e^{U/\gamma+1}$ compared to the standard case (Section 1.3); when $U = 0$ it reduces to the constant e, which is absorbed into the normalisation of the potentials.

Equations for the potentials

Substituting (34) into the Euler–Lagrange–Bellman system of the gSB problem, one can verify that η^{fw} and η^{bw} satisfy forward and backward equations governed by the *Fokker–Planck operator with potential U*:

$$\mathrm{FP}^U(\eta) := \gamma \Delta \eta + \nabla \cdot (\eta \nabla U). \tag{35}$$

Specifically,

$$\begin{aligned} \partial_t \eta^{\mathrm{fw}} &= \mathrm{FP}^U(\eta^{\mathrm{fw}}), \quad \eta^{\mathrm{fw}}\big|_{t=0} = \eta_0^{\mathrm{fw}}, \\ -\partial_t \eta^{\mathrm{bw}} &= \mathrm{FP}^U(\eta^{\mathrm{bw}}), \quad \eta^{\mathrm{bw}}\big|_{t=1} = \eta_1^{\mathrm{bw}}. \end{aligned} \tag{36}$$

The forward equation for η^{fw} is a standard Fokker–Planck equation (diffusion + advection towards low-U regions); the backward equation for η^{bw} is the same operator run in reverse time. The stationary measure of FP^U is $\pi \propto e^{-U/\gamma}$, which links back to the reference measure introduced in (27).

6.4 Generalized Schrödinger System and Sinkhorn Iteration

Marginal conditions

Evaluating the density (34) at $t = 0$ and $t = 1$:

$$\rho_0(x) = \eta_0^{\mathrm{fw}}(x) \cdot \eta^{\mathrm{bw}}(0,x) \cdot e^{U(x)/\gamma+1}, \tag{37}$$
$$\rho_1(x) = \eta^{\mathrm{fw}}(1,x) \cdot \eta_1^{\mathrm{bw}}(x) \cdot e^{U(x)/\gamma+1}. \tag{38}$$

These are the unknowns $(\eta_0^{\mathrm{fw}}, \eta_1^{\mathrm{bw}})$ that must be found. Denote by P_t^U the semigroup of FP^U: $\eta^{\mathrm{fw}}(t,\cdot) = P_t^U[\eta_0^{\mathrm{fw}}]$ and $\eta^{\mathrm{bw}}(0,\cdot) = P_1^U[\eta_1^{\mathrm{bw}}]$. The system (37)–(38) becomes

$$\eta_0^{\mathrm{fw}}(x) \cdot P_1^U[\eta_1^{\mathrm{bw}}](x) \cdot e^{U(x)/\gamma+1} = \rho_0(x), \tag{39}$$

$$P_1^U[\eta_0^{\mathrm{fw}}](x) \cdot \eta_1^{\mathrm{bw}}(x) \cdot e^{U(x)/\gamma+1} = \rho_1(x). \tag{40}$$

Generalized Sinkhorn iteration

Equations (39)–(40) are solved by the same alternating-projection structure as Algorithm 1, now with the $e^{U/\gamma+1}$ correction folded in:

Algorithm 2 Generalized Hopf–Cole Sinkhorn (gSB with potential U)

1: Initialize $\eta_0^{\mathrm{fw}(0)} = \mathbf{1}$, $\eta_1^{\mathrm{bw}(0)} = \rho_1\, e^{-U/\gamma-1}$
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: $\quad \eta_0^{\mathrm{fw}(k+1)} \leftarrow \dfrac{\rho_0}{P_1^U[\eta_1^{\mathrm{bw}(k)}] \cdot e^{U/\gamma+1}}$ *(enforce $\rho(0) = \rho_0$)*
4: $\quad \eta_1^{\mathrm{bw}(k+1)} \leftarrow \dfrac{\rho_1}{P_1^U[\eta_0^{\mathrm{fw}(k+1)}] \cdot e^{U/\gamma+1}}$ *(enforce $\rho(1) = \rho_1$)*
5: **end for**

Each step requires one application of P_1^U, which solves equation (36) for one unit of time. The convergence guarantee of the standard Sinkhorn extends to this setting provided P_t^U is positivity-preserving and mass-conserving (both of which hold for FP^U with Neumann BCs on bounded domains).

Comparison with the standard SB ($U = 0$)

	Standard SB ($U = 0$)	**Generalized SB** ($U \neq 0$)
Reference measure	uniform	$\pi \propto e^{-U/\gamma}$
Potential equations	heat eq. $\partial_t \varphi = \gamma \Delta \varphi$	$\partial_t \eta^{\mathrm{fw}} = \mathrm{FP}^U(\eta^{\mathrm{fw}})$
Density factorisation	$\rho = \varphi\hat{\varphi}$	$\rho = \eta^{\mathrm{fw}}\eta^{\mathrm{bw}} e^{U/\gamma+1}$
Semigroup P_t	Neumann heat kernel (DCT-II)	FP semigroup (PDE solve)
Sinkhorn update	divide by $P_1[\,\cdot\,]$	divide by $P_1^U[\,\cdot\,] \cdot e^{U/\gamma+1}$

Remark 2. *The implementation cost increases relative to the $U = 0$ case because P_t^U no longer diagonalises in the DCT-II basis (unless U is quadratic, in which case a Mehler-kernel formula applies). In general, each application of P_1^U requires solving the Fokker–Planck PDE (36) numerically over $[0,1]$, raising the per-iteration cost from $\mathcal{O}(N^d \log N)$ to $\mathcal{O}(N^d \cdot N_t)$.*

7 Summary and Outlook

We have implemented a clean, fully vectorized solver for the Schrödinger Bridge problem based on the Hopf–Cole factorization and Sinkhorn iteration. The key features of the implementation are:

- **Exact mass conservation** by construction (DCT-II $k = 0$ mode).
- **Analytically exact continuity equation** — no time-stepping error in the density trajectory once potentials have converged.

- **Scalable to d dimensions** via separable DCT, with $\mathcal{O}(N^d \log N)$ cost per iteration.
- **Validated against closed-form Gaussian SB solution** with L^∞ errors below 10^{-6} in 1D.

Possible extensions include:

- Non-Gaussian or non-isotropic marginals (e.g. two-hump to one-hump).
- Newton acceleration of the Sinkhorn fixed-point (see `newton_logPotentials.py`).
- Coupling with the dynamic ADMM solver for the Fokker–Planck formulation at $\varepsilon = 0$ (Benamou–Brenier OT).

A Analytical Gaussian SB Derivation

For $\rho_0 = \mathcal{N}(\mu_0, \sigma_0^2)$ and $\rho_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ on $\mathbb{R}$ with diffusion $\varepsilon/2$, we seek Gaussian Hopf–Cole potentials

$$\varphi_0(x) \propto \exp\Big(-\tfrac{(x-m_0)^2}{2\alpha^2}\Big), \tag{41}$$

$$\hat{\varphi}_1(x) \propto \exp\Big(-\tfrac{(x-n_1)^2}{2\beta^2}\Big). \tag{42}$$

Their heat-evolved counterparts are also Gaussian: $P_t[\varphi_0] \propto \mathcal{N}(m_0, \alpha^2 + \varepsilon t)$ and $P_{1-t}[\hat{\varphi}_1] \propto \mathcal{N}(n_1, \beta^2 + \varepsilon(1-t))$.

The marginal conditions $\varphi_0 P_1[\hat{\varphi}_1] = \rho_0$ and $P_1[\varphi_0]\hat{\varphi}_1 = \rho_1$ give, matching precision (inverse variance):

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2 + \varepsilon} = \frac{1}{\sigma_0^2}, \tag{43}$$

$$\frac{1}{\alpha^2 + \varepsilon} + \frac{1}{\beta^2} = \frac{1}{\sigma_1^2}, \tag{44}$$

and matching the mean:

$$\begin{pmatrix} 1/\alpha^2 & 1/(\beta^2+\varepsilon) \\ 1/(\alpha^2+\varepsilon) & 1/\beta^2 \end{pmatrix} \begin{pmatrix} m_0 \\ n_1 \end{pmatrix} = \begin{pmatrix} \mu_0/\sigma_0^2 \\ \mu_1/\sigma_1^2 \end{pmatrix}. \tag{45}$$

The nonlinear system for (α^2, β^2) is solved numerically with `scipy.optimize.fsolve`; the linear system for (m_0, n_1) is solved analytically by Cramer's rule.